THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **Some Applications of Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A storm breaks a tree at a height of 9 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 60° with the ground. The original height of the tree is: 1

(a) 18 m **(b)** 15 m **(c)** $9 + 6\sqrt{3}$ m **(d)** $6\sqrt{3}$ m

2. The string of a kite is 46 m long. It is taut and makes an angle of 60° with the level ground. The height of the kite above the ground is: 1

(a) 23 m **(b)** $46\sqrt{3}$ m **(c)** $\frac{23\sqrt{3}}{2}$ m **(d)** $23\sqrt{3}$ m

3. A statue 18 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. The height of the pedestal is: 1

(a) 18 m **(b)** $18\sqrt{3}$ m **(c)** $9 + 9\sqrt{3}$ m **(d)** $6\sqrt{3}$ m

4. The height of a tower is h and the angle of elevation of its top is 45°. The distance of the observer from the foot of the tower is: 1

(a) $2h$ **(b)** $\sqrt{2}\,h$ **(c)** $\frac{h}{2}$ **(d)** h

5. A boy whose eyes are 4 m above the ground stands 30 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) 35 m **(b)** $34\sqrt{3}$ m **(c)** 30 m **(d)** 34 m

6. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 24 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) $8\sqrt{3}$ m **(b)** $12\sqrt{3}$ m **(c)** 12 m **(d)** $24\sqrt{3}$ m

7. Two points on level ground lie on opposite sides of a tower of height $30\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. The distance between the two points is: 1

(a) 120 m **(b)** 60 m **(c)** $120\sqrt{3}$ m **(d)** 240 m

8. A storm breaks a tree at a height of 14 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The distance from the foot of the tree to the point where its top touches the ground is: 1

(a) 14 m **(b)** $14\sqrt{3}$ m **(c)** $\frac{14\sqrt{3}}{3}$ m **(d)** 28 m

9. A storm breaks a tree at a height of 12 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 60° with the ground. The original height of the tree is: 1

(a) 24 m **(b)** 20 m **(c)** $12 + 8\sqrt{3}$ m **(d)** $8\sqrt{3}$ m

10. The string of a kite is 52 m long. It is taut and makes an angle of 60° with the level ground. The height of the kite above the ground is: 1

(a) $52\sqrt{3}$ m **(b)** 26 m **(c)** $13\sqrt{3}$ m **(d)** $26\sqrt{3}$ m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. From the top of a tower 18 m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 45° and 60°. Find the distance between the two objects. **2**

12. From the top of a building 24 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. Find the height of the tower. **2**

13. A boy whose eyes are 1 m above the ground stands 21 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 60°. Find the height of the building. **2**

14. From the top of a tower 30 m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 45° and 60°. Find the distance between the two objects. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 18 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. **3**

16. A building is $14\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). **3**

17. A flagstaff stands on the top of a building 15 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. **3**

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 20 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. **3**

18. A building is $16\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). **3**

OR

A flagstaff stands on the top of a building 20 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two vertical poles of equal height stand on opposite sides of a road 220 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). **5**

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 10 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. Find the height of the tower and the initial distance of the point from its foot (in this order). **5**

20. Two vertical poles of equal height stand on opposite sides of a road 260 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). **5**

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 12 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. Find the height of the tower and the initial distance of the point from its foot (in this order). **5**

END OF QUESTION PAPER — Total: 40 Marks